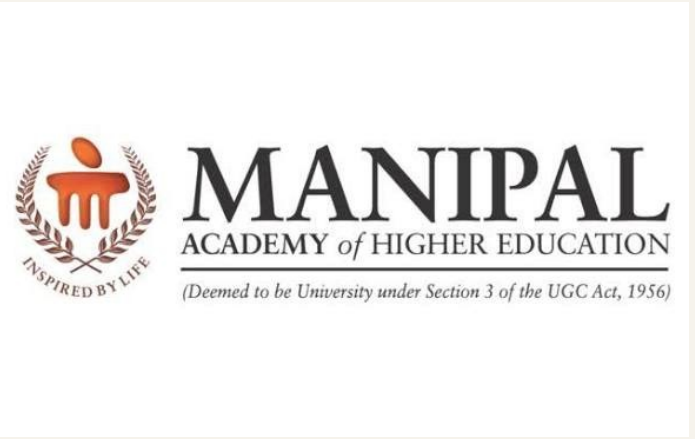


INDUSTRIAL TRAINING PRESENTATION

Remote Sensing and GIS Using Python

Project 1 Drought severity in Jalna District, 2023
Project 2 Urban heat in Delhi, April–June 2024

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OVERVIEW

Two projects, one way of working

PROJECT 1

Jalna District, Maharashtra

Where was vegetation stressed in 2023, compared with 2015–2023?

MODIS MOD13Q1 Version 6.1
250 m NDVI, every 16 days
Vegetation Condition Index
Mean VCI for the district: 0.488

PROJECT 2

National Capital Territory of Delhi

Where was the land surface hottest in April–June 2024?

Landsat-8 Collection 2 Level-2
30 m, cloud cover below 10%
NDVI, NDBI and surface temperature
Mean land surface temperature: 46.66°C

Same path for both projects

Administrative boundary → process the satellite record in Google Earth Engine → map layout in QGIS 3.40

How the drought map was made

1

Satellite record

MODIS MOD13Q1 Version 6.1. NDVI at 250 m, 16-day composites, from 2015 to 2023. Values were scaled by 0.0001.

2

Vegetation Condition Index

VCI = (NDVI 2023 − NDVI minimum) / (NDVI maximum − NDVI minimum). The 2023 value is the yearly mean.

3

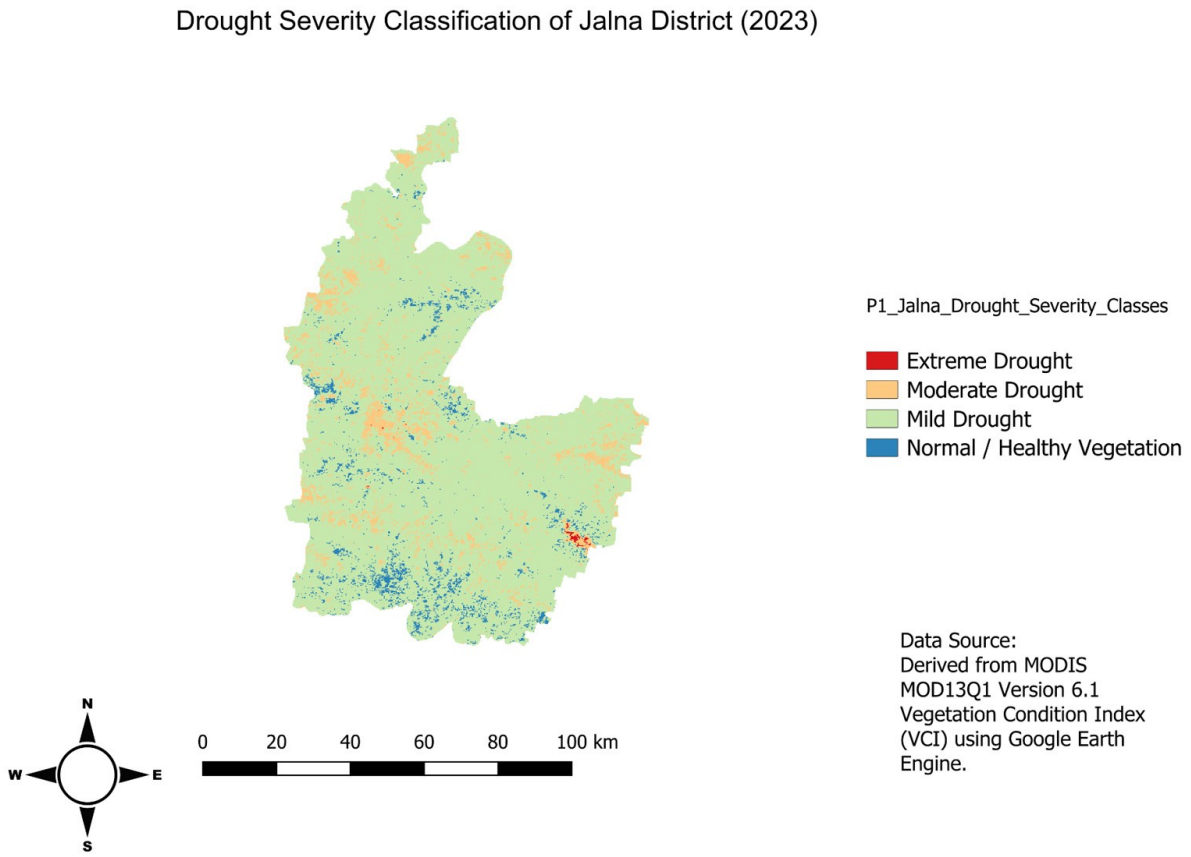
Four classes, then area

Each pixel was classed, clipped to the FAO GAUL boundary, and counted in Google Earth Engine. Layouts were made in QGIS 3.40.

Class breaks: 0.00–0.20 extreme · 0.20–0.40 moderate · 0.40–0.60 mild · 0.60–1.00 healthy

A district mean of 0.488 falls in the mild class.

Most of Jalna was in mild drought



Jalna District, 2023

0.488
Mean Vegetation Condition Index

6,821.66 km²
Mild drought · 87.69%

652.14 km²
Moderate drought

8.55 km²
Extreme drought

296.53 km²
Healthy vegetation

The four classes sum to 7,778.88 km². Stress was widespread. The extreme class was small.

How the heat map was made

Landsat-8 Collection 2 Level-2

April–June 2024 · cloud cover below 10% · 30 m · median composite · FAO GAUL clip

NDVI

Vegetation

(NIR − Red) / (NIR + Red)

Bands 5 and 4

NDBI

Built-up

(SWIR − NIR) / (SWIR + NIR)

Bands 6 and 5

LST

Temperature

(ST_B10 × 0.00341802 + 149) − 273.15

Degrees Celsius

Heat zones used for the area table

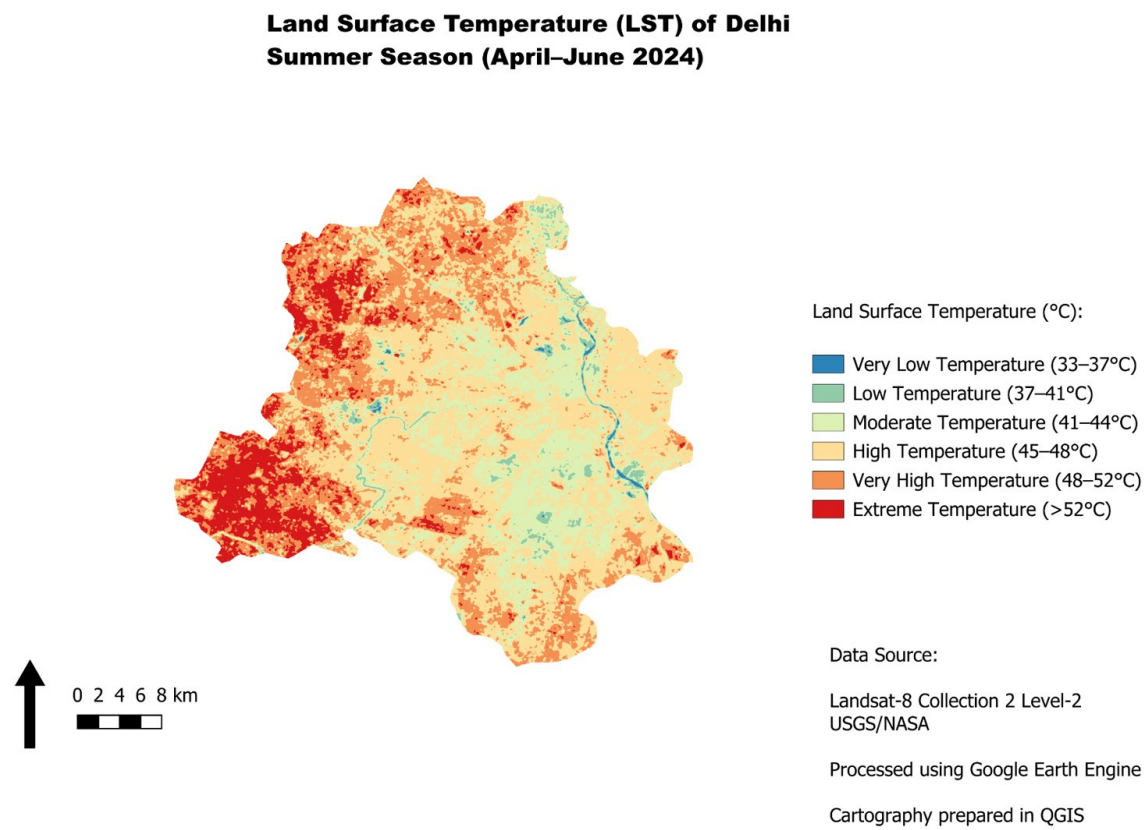
Below 40°C
Low

40–45°C
Moderate

45–50°C
High

Above 50°C
Very high · hotspot

A hot surface, with a cooler river corridor



Delhi, April–June 2024

33.10°C
Minimum

57.71°C
Maximum

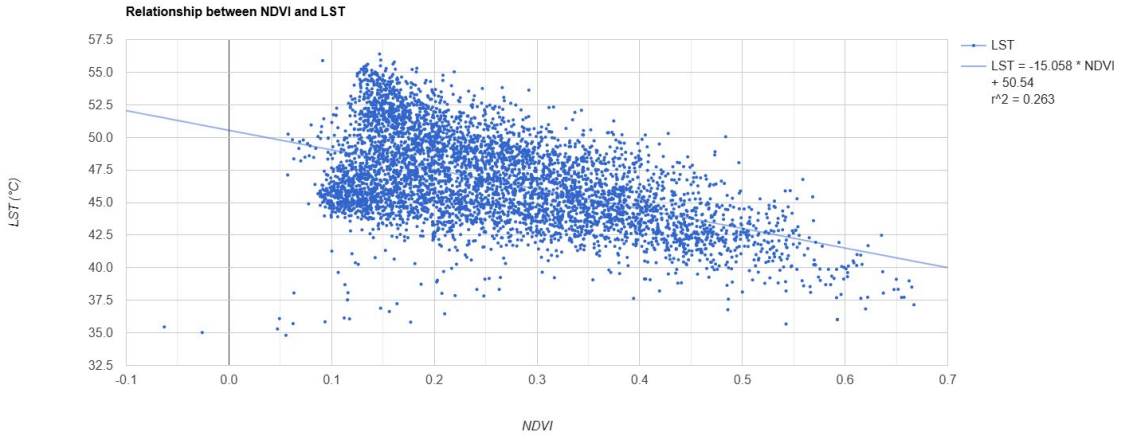
46.66°C
Mean · s.d. 3.36°C

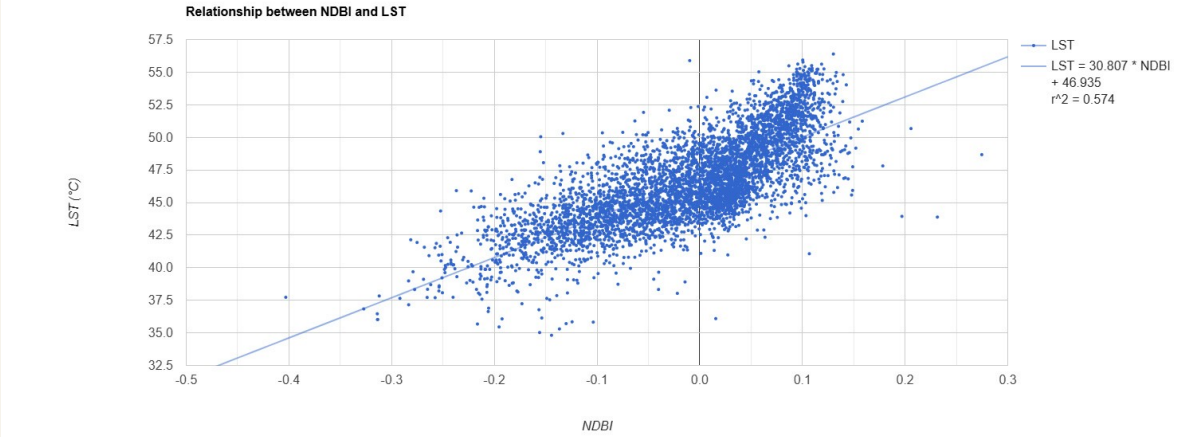
259.16 km²
Above 50°C

17.25%
Of the classified 1,502.20 km²

Cooler near the Yamuna, parks and woodland.
Hotter in the west and south-west.

Greener pixels are cooler. Built-up pixels are hotter.





NDVI and temperature

LST = −15.058 × NDVI + 50.54

r^2 = 0.263 · the line slopes down

NDBI and temperature

LST = 30.807 × NDBI + 46.935

r^2 = 0.574 · the line slopes up

The scatter is wide, especially for NDVI. In this summer composite the built-up index lines up more closely with surface temperature. These fits describe the plotted pixels. They do not, by themselves, prove cause.

CLOSING

What these maps can support

Jalna, 2023

Mild drought covers 87.69% of the classified area. Extreme drought is 8.55 km^2. The mean VCI is 0.488.

Delhi, summer 2024

Land surface temperature runs from 33.10°C to 57.71°C. The area above 50°C is 17.25% of the classified surface.

How to read them

Both are satellite screening maps. They were not checked against field measurements, rainfall, or air temperature.

Thank you.

I can take questions on the data, the class breaks, or the maps.